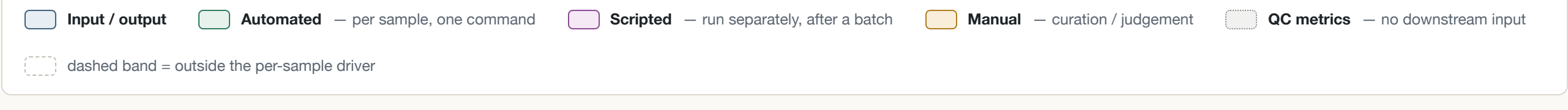
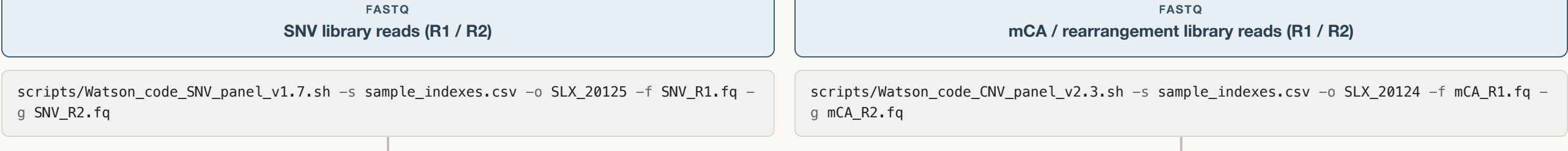


TETRS-seq bioinformatic pipeline

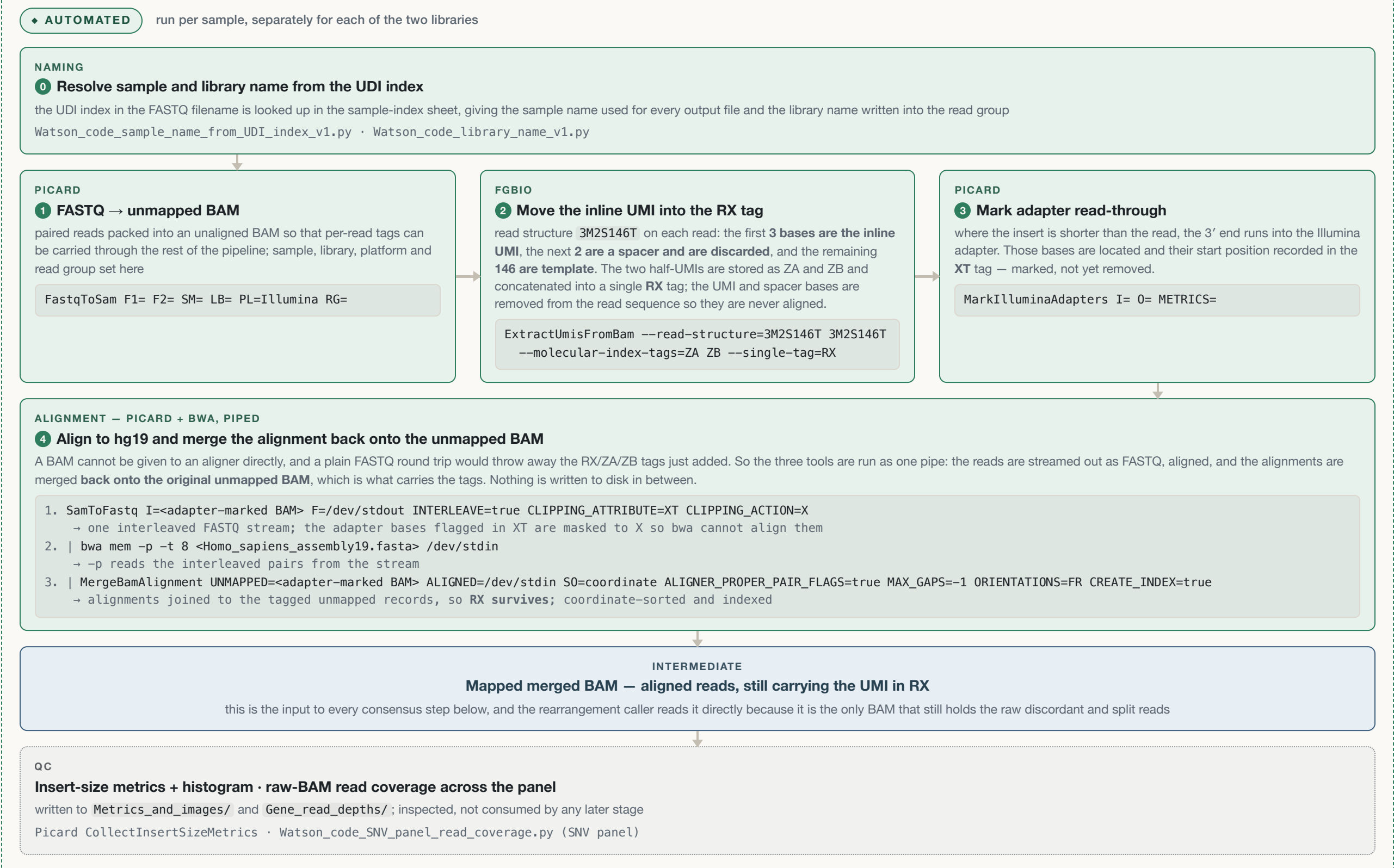
Two capture panels are processed independently from **separate FASTQ libraries**. Both begin with the same read-preparation and alignment front end, then diverge. Per-sample stages are **automated**; the cross-sample and longitudinal stages are **triggered separately** once a batch of samples has been processed — most are scripts in this repository, and only a few are **manual**.



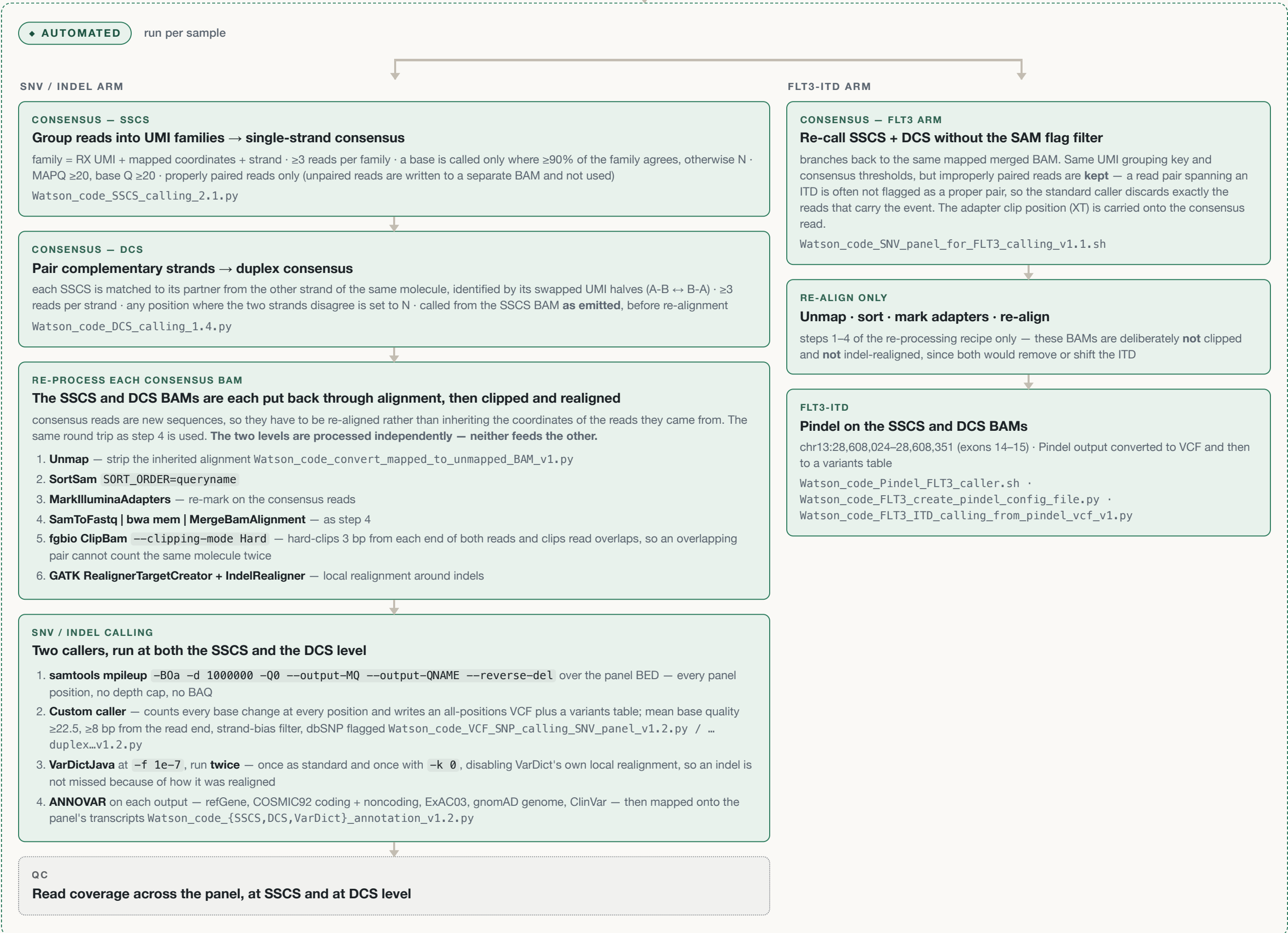
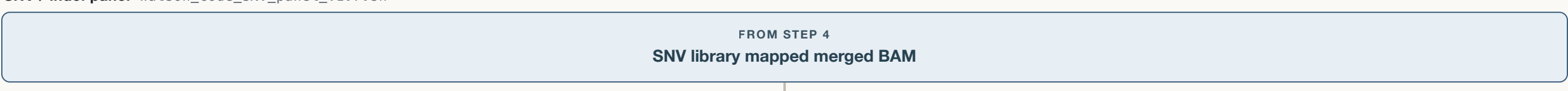
Read preparation and alignment steps 0–4 – the same code in both panel scripts
TWO SEPARATE RUNS – ONE COMMAND PER PANEL, EACH GIVEN THAT PANEL'S OWN FASTQ PAIR



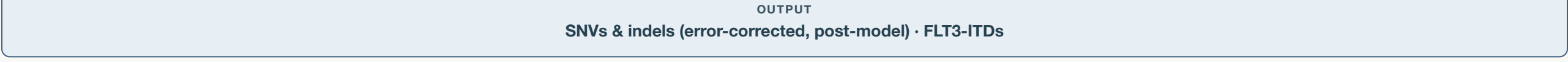
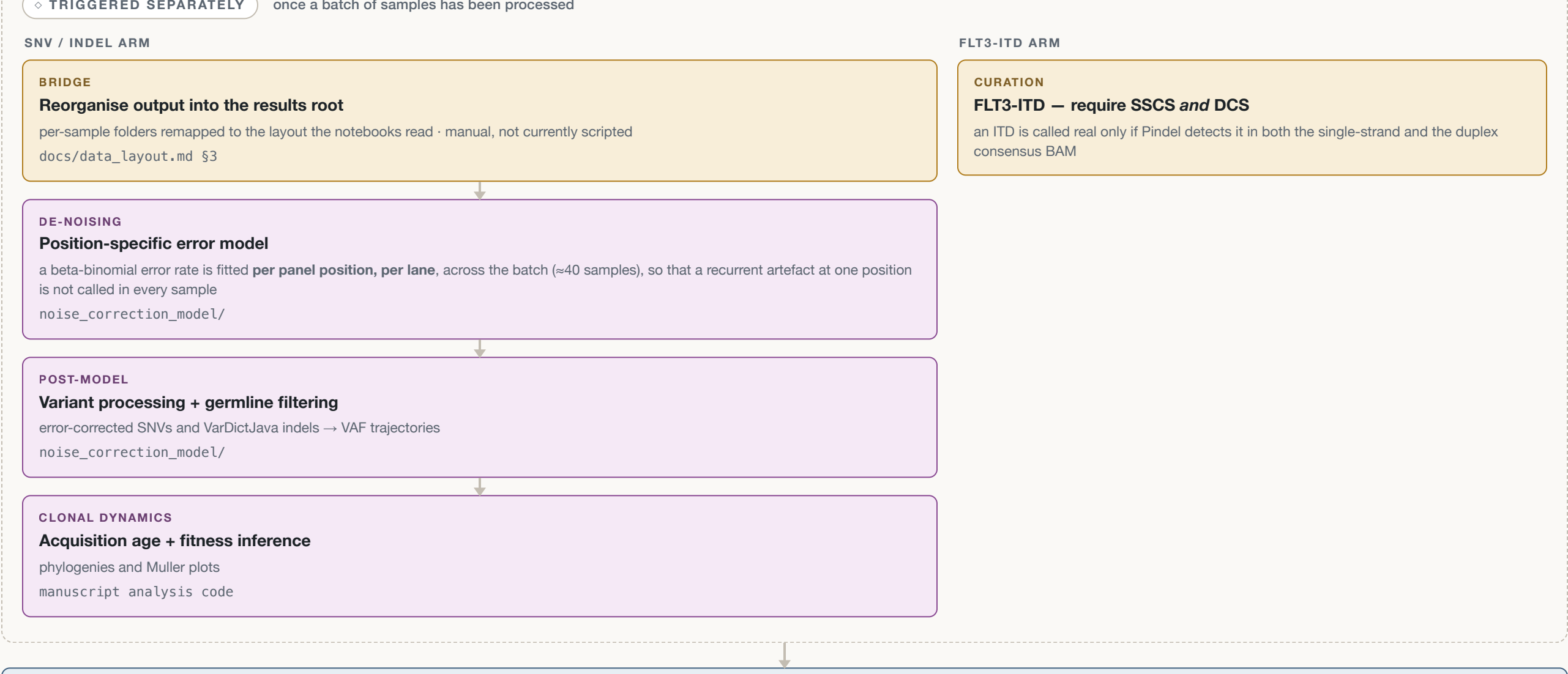
EACH COMMAND BEGINS WITH THE IDENTICAL STEPS 0-4 BELOW, ON ITS OWN LIBRARY



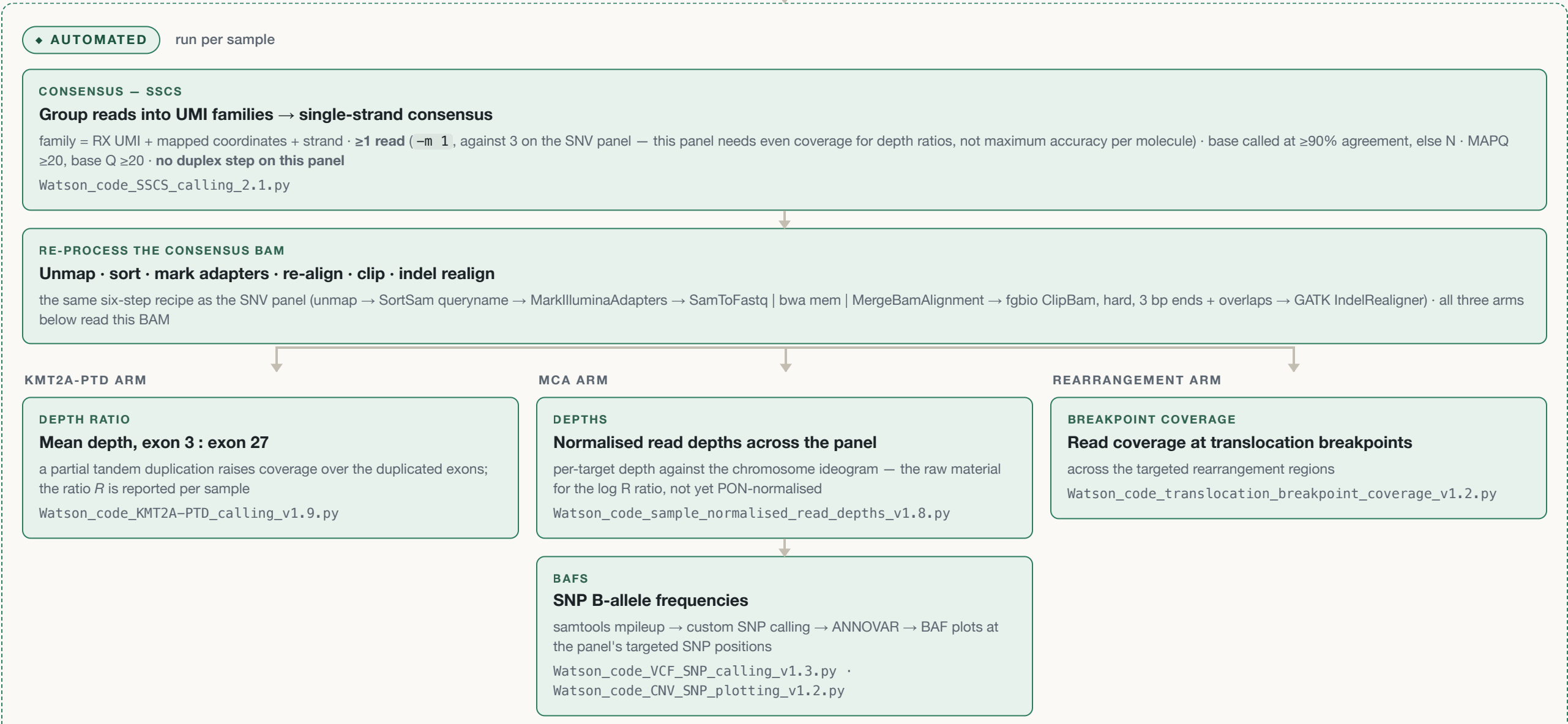
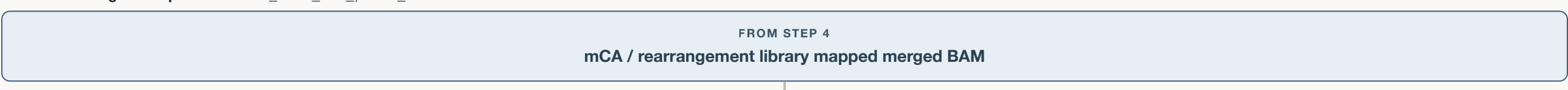
SNV / indel panel Watson_code_SNV_panel_v1.7.sh



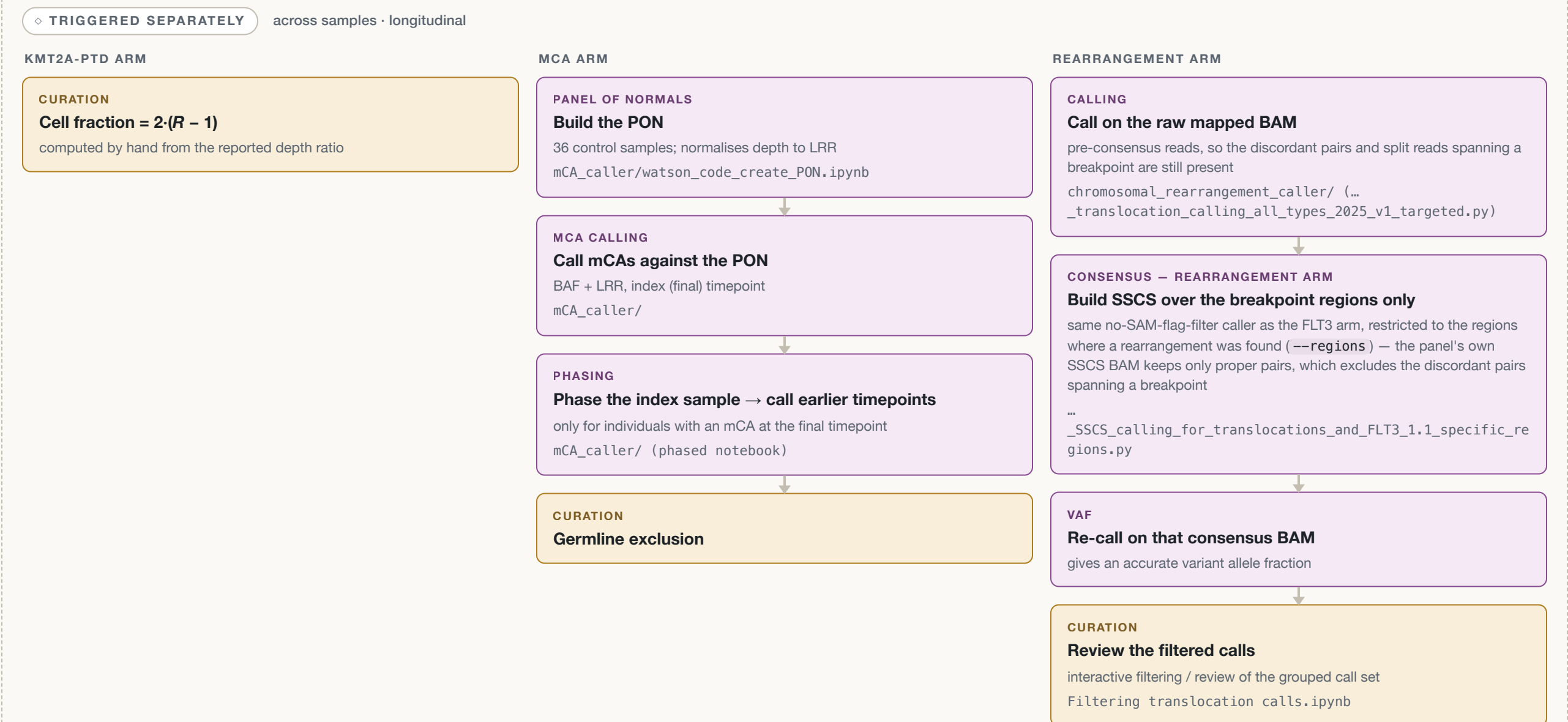
TRIGGERED SEPARATELY once a batch of samples has been processed



mCA / rearrangement panel Watson_code_CNV_panel_v2.3.sh



TRIGGERED SEPARATELY across samples · longitudinal



NOTES

Paired-end reads from two independent capture panels — an SNV/indel panel and an mCA/rearrangement panel, each sequenced as a separate library — are processed from raw FASTQ. Green stages are fully automated and run per sample by the pipeline driver (one command each); purple stages are scripts and notebooks in this repository, triggered separately once a batch of samples has been processed; amber stages are manual; dotted grey boxes are QC metrics that are inspected but not consumed downstream.

Read preparation and alignment (steps 0–4) are identical for both libraries. The sample and library names are resolved from the UDI index in the FASTQ filename; the reads are converted to an unaligned BAM (*FastqToSam*) so that per-read tags can be carried through the pipeline. Each read begins with a 3 bp inline UMI, a 2 bp spacer and 146 bp of template (read structure 3M2S146T); the UMI bases are moved into the read's RX tag and removed from the sequence (*ExtractUmisFromBam*), and adapter read-through at the 3' end is marked in the XT tag (*MarkIlluminaAdapters*). Alignment is run as a single pipe — *SamToFastq* streams the reads out as interleaved FASTQ with the adapter bases masked so they are not aligned, *bwa mem* aligns them to hg19, and *MergeBamAlignment* merges the alignments back onto the unaligned BAM — so that the UMI and all other tags survive alignment. The resulting mapped merged BAM is the input to every consensus step, and is also the BAM the rearrangement caller reads, because it is the only one that still contains the raw discordant and split reads.

Consensus calling. Reads are grouped into families sharing a UMI, mapping coordinates and strand, and a single-strand consensus (SSCS) is called from each family, a base being called only where ≥90% of the family agrees. The minimum family size is 3 for the SNV panel and 1 for the mCA panel. For the SNV panel each SSCS is then paired with its complementary-strand partner — the same molecule read from the other strand, identified by its swapped UMI halves — to give a duplex consensus (DCS), with positions of strand disagreement set to N; the duplex consensus is called from the single-strand consensus BAM as emitted. Because consensus reads are new sequences, each consensus BAM is put back through the same alignment round trip and then hard-clipped (3 bp from each read end and across read overlaps) and realigned around indels. The SSCS and DCS BAMs are processed independently, each feeding only its own variant calling. Variants are called from a full-depth, unclipped *samtools mpileup* by a custom caller that records every base change at every panel position, and independently by *VarDictJava*, run twice — with and without its own local indel realignment — with ANNOVAR annotation of each output.

FLT3-ITD detection branches back to the mapped merged BAM and builds its own pair of SSCS and DCS BAMs, using the same UMI grouping and consensus thresholds but without the SAM flag filter, so that the improperly paired and soft-clipped reads carrying an ITD are retained; these are re-aligned but neither clipped nor indel-realigned, and Pindel is run over FLT3 exons 14–15.

Cross-sample stages. After a batch of samples has been processed, per-sample output is reorganised manually into the results-root layout the analysis notebooks read; the position-specific error model is fitted across the batch (≈40 samples per lane) and its output is filtered for germline and artefactual variants to give variant-allele-fraction trajectories, from which acquisition age, fitness, phylogenies and Muller plots are derived; the mosaic chromosomal alteration (mCA) caller builds a panel of normals from 36 controls, calls mCAs, and — where an mCA is present at the final timepoint — phases the index sample and calls at earlier timepoints; chromosomal rearrangements are called on the raw BAM and re-called on the SSCS BAM for accurate variant-allele-fraction estimation. The manual steps are that reorganisation, the requirement that an FLT3-ITD be detected in both SSCS and DCS, the KMT2A-PTD cell fraction, exclusion of germline events from the mCA calls, and review of the filtered rearrangement calls; every other stage shown is code in this repository.